Lowering the Mills Exponent by Quantifying Sieve Exceptional Sets

An extension of harmonic branching to $c > 43/41$

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Abstract

Let $\mathcal{W}(c)$ consist of the real numbers $A > 1$ for which $\lfloor A^{c^n} \rfloor$ is prime for every $n \geq 1$. The harmonic branching manuscript supplied with this note obtains $c > 15/13$ from a short-interval prime number theorem. We give a more economical analytic input: a positive proportion of the expected primes, with sufficiently small exceptional measure. We quantify the finite sieve argument of Li, *Primes in almost all short intervals*, arXiv:2407.05651v6, with the free accuracy parameter $\mathcal{Q}_D(X) = \exp(D(\log\log X)^2)$. The Vinogradov–Korobov zero-free region supplies every fixed power of $\mathcal{Q}_D^{-1}$ needed in the argument; all moment orders, sieve parameters, and combinatorial depths are fixed before X increases. The resulting estimate, combined with harmonic branching, gives $\mathcal{W}(c) \neq \varnothing$ for every $c > 43/41 = 1.048780\ldots$, and in fact uncountability and unboundedness. The numerical region inequalities in Li's finite sieve decomposition are an explicit external input, not independently recomputed in this note. No Riemann hypothesis or density hypothesis is used. The sharper potential targets 22/21 and 24/23 are recorded separately: their required quantitative sieve estimates are not proved here. Applying the same accuracy argument to Jia's published 1996 proof already gives the independent range $c > 20/19$.

1 Result, scope, and dependencies

Put

$$\mathcal{W}(c) = \{A > 1 : \lfloor A^{c^n} \rfloor \in \mathcal{P} \text{ for every } n \geq 1\}, \qquad J_c(m) = [m^c, (m+1)^c - 1).$$

Here c is the exponent parameter, and A is the representing constant. Neither the least element of $\mathcal{W}(3)$ nor its numerical value is changed.

Theorem 1.1 (quantitative extension)**.** *Using the finite sieve decomposition and numerical region bounds in Li [2], $\mathcal{W}(c)$ is uncountable, unbounded, Lebesgue-null and nowhere dense for every real*

$$\boxed{c > \frac{43}{41}}.$$

More generally, if $C_n = c_1 \cdots c_n$ and $43/41 < \inf_n c_n \leq \sup_n c_n < \infty$, the same assertions hold for $\{A > 1 : \lfloor A^{C_n} \rfloor \in \mathcal{P} \text{ for all } n \geq 1\}$.

The qualification in the statement identifies the exact imported component. Li's paper is a preprint, and its positive sieve margin rests on explicit numerical integral inequalities. We use those inequalities as stated; we do not supply a new interval-arithmetic certificate for them. The new analytic work here concerns the uniform exceptional-set estimate and its transfer to the prime tree. Thus this is a research proof extension with documented dependencies, not a claim of independent verification of every calculation in the cited literature.

Theorem 1.2 (version using published sieve inputs)**.** *Using Jia's published finite sieve argument [5], the conclusions of Theorem 1.1 hold for $c > 20/19$ and for bounded variable exponent sequences whose infimum exceeds $20/19$. This version does not require the improved regions or numerical inequalities of Li's preprint.*

Input or target	interval exponent	Mills boundary	Status in this note
Supplied manuscript	2/15	15/13	Baseline argument
Jia [5]	1/20	20/19	Quantitative extension of published sieve proof
Li I [2]	2/43	43/41	Quantitative extension proved below, using its finite sieve margin
Li II [3]	1/22	22/21	Further quantitative verification needed
Li III [4]	1/24	24/23	Author-site manuscript; further quantitative verification needed

All displayed boundaries in this table concern strict inequalities. In particular, $c = 43/41$ is not included in Theorem 1.1. A convenient covered example is $c = 21/20 = 1.05$, since

$$1 - \frac{1}{21/20} = \frac{1}{21} > \frac{2}{43}.$$

The reduction is

$$\frac{15}{13} - \frac{43}{41} = \frac{56}{533} = 0.105065\ldots.$$

Measured by the excess over 1, it replaces 2/13 by 2/41.

2 Why fixed logarithmic estimates cannot simply be substituted

For each separately fixed B, Li's stated theorem supplies exceptional measure $O_B(X(\log X)^{-B})$. The harmonic construction needs a uniform estimate of the form

$$X \exp(-D(\log\log X)^2).$$

The first assertion alone does not imply the second. For example,

$$r(X) = \exp(-(\log\log X)^{3/2})$$

is $O_B((\log X)^{-B})$ for every fixed B, but exceeds $\exp(-D(\log\log X)^2)$ eventually for every fixed $D > 0$.

This distinction has a concrete tree interpretation. At level n, $\log Y_n$ is comparable to c^n, and repeatedly retaining a prime proportion of size $1/\log Y_n$ loses

$$\prod_{j<n} \frac{1}{\log Y_j} = \exp\left(-\frac{\log c}{2} n^2 + O(n)\right).$$

A fixed logarithmic exceptional rate gives only $\exp(-Bn\log c + O(1))$, which eventually exceeds this retained mass. The analytic proof must therefore be made quantitative before it is combined with the tree argument.

For $D > 0$ define, for sufficiently large X,

$$\mathcal{Q}_D(X) = \exp(D(\log\log X)^2), \qquad \mathcal{L}(X) = \frac{(\log X)^{1/3}}{(\log\log X)^{1/3}}.$$

For any fixed $D, K, d, \nu > 0$,

$$(\log X)^d = \mathcal{Q}_D(X)^{o(1)}, \quad \mathcal{Q}_D(X)^K = X^{o(1)}, \quad \mathcal{Q}_D(X)^K e^{-\nu\mathcal{L}(X)} = o(\mathcal{Q}_D(X)^{-d}). \tag{1}$$

The order of choices below is essential: fix the interval exponent, then fix all sieve and moment parameters, then choose the finite accuracy powers, and finally let X increase.

3 A quantitative prime-polynomial estimate

Lemma 3.1 (uniform prime-polynomial bound)**.** *Fix $\delta > 0$, $M_0 > 0$, and a bounded real interval of values of σ. There is $a > 0$ such that, whenever X is sufficiently large, $X^\delta \le u \le v \le 2u \ll X$, and $|t| \le X^{M_0}$,*

$$\sum_{u<n\le v} \frac{\Lambda(n)}{n^{\sigma+it}} = \int_u^v y^{-\sigma-it}\,dy + O_{\delta,M_0,\sigma}\left(u^{1-\sigma}e^{-a\mathcal{L}(X)}\right).$$

The constants are uniform when σ stays in the indicated compact interval. Consequently

$$\left|\sum_{u<n\le v} \frac{\Lambda(n)}{n^{\sigma+it}}\right| \ll u^{1-\sigma}\left(\frac{1}{1+|t|} + e^{-a\mathcal{L}(X)}\right).$$

The analogous upper bound for primes with coefficient 1 *holds, with at most a fixed logarithmic factor.*

Proof. Let $s = \sigma + it$. Apply truncated Perron's formula to $-\zeta'(s+w)/\zeta(s+w)$ with difference kernel

$$\frac{v^w - u^w}{w}.$$

Take the initial line with $\Re(s+w) > 1$, and choose a truncation height X^M with fixed M sufficiently large relative to M_0 and the allowed range of σ. The endpoint convention for Perron's formula changes the normalized sum by $O(u^{-\sigma}\log X)$, which is absorbed by the asserted error.

The Vinogradov–Korobov zero-free region permits a shift to $\Re(s+w) = 1-\eta$, where

$$\eta = b_0(\log X)^{-2/3}(\log\log X)^{-1/3}$$

and $b_0 > 0$ is sufficiently small. Decreasing b_0 handles bounded heights as well. In this narrower zero-free strip, $-\zeta'/\zeta(z) - (z-1)^{-1}$ has a fixed power of $\log X$ as an upper bound. The only pole crossed is at $w = 1-s$, and its residue is

$$\frac{v^{1-s} - u^{1-s}}{1-s} = \int_u^v y^{-s}\,dy.$$

The difference kernel is regular at $w = 0$, so it creates no additional residue there. The shifted vertical integral is

$$\ll u^{1-\sigma-\eta}(\log X)^{O(1)} \ll u^{1-\sigma}e^{-a\mathcal{L}(X)},$$

because $\log u \ge \delta \log X$. The horizontal integrals and Perron truncation error are smaller on increasing M. This proves the expansion.

For the main integral, writing $y = ue^z$ shows that its size is $O(u^{1-\sigma}/(1+|t|))$, uniformly for $0 \le \log(v/u) \le \log 2$. Removing prime powers costs at most a fixed logarithmic factor times $u^{1/2-\sigma} + u^{-\sigma}$, which is negligible here. Partial summation removes the remaining $\log p$ weight. □

Corollary 3.2. *For every fixed $D, K > 0$, the prime polynomials in Lemma 3.1 satisfy*

$$\left|\sum_{u<n\le v} \Lambda(n)n^{-\sigma-it}\right| \ll u^{1-\sigma}\mathcal{Q}_D(X)^{-K}$$

throughout $\mathcal{Q}_D(X)^{K+1} \le |t| \le X^{M_0}$, for sufficiently large X. The low-frequency main integral is retained rather than discarded.

Lemma 3.3 (fixed products of large primes)**.** *The same conclusion, with a possible fixed logarithmic loss, holds for Dirichlet polynomials formed from a fixed number of prime variables $p_j \ge X^\delta$, with fixed order, range, or monomial constraints, provided the selected prime variable has no arbitrary oscillating coefficient. It continues to hold after convolution with bounded sieve coefficients in other variables, with fixed divisor and logarithmic losses.*

Proof. Fix all but one prime variable and let q be their product. The allowed values of the remaining prime lie in a bounded union of intervals inside $[M/q, 2M/q]$, subject to a lower bound X^{δ}. Use Lemma 3.1 on that variable. Its error, multiplied by $q^{-\sigma}$, is $O(M^{1-\sigma}q^{-1}e^{-a\mathcal{L}(X)})$. The sum of $1/q$ over a fixed number of primes in $[X^{\delta}, O(X)]$ is bounded in terms of δ and the number of variables, by Mertens' estimate. Fixed logarithmic weights add only fixed logarithmic factors.

One can fix the other sieve variables first as well. Bounded sieve coefficients have harmonic absolute sums bounded by fixed logarithmic powers; grouping a fixed number of variables gives a fixed divisor bound. These factors are absorbed using (1). This statement does not assert cancellation for arbitrary coefficients on the prime variable itself. □

4 Quantifying the finite sieve argument

We use Li's finite decomposition at the short-interval exponent 2/43. Its combinatorial regions and final numerical integrals are unchanged. The issue is the analytic error budget.

Proposition 4.1 (accuracy parameter in the finite sieve)**.** *Fix a sufficiently small $\varepsilon > 0$ in the argument of [2], and put*

$$\eta = \frac{1}{2}X^{-41/43+\varepsilon}.$$

For every fixed $D > 0$, there exist X_0 and $\kappa > 0$ such that

$$\operatorname{meas}\left\{x \in [X, 2X] : \pi(x+\eta x) - \pi(x) < \kappa\frac{\eta x}{\log X}\right\} \le X\,\mathcal{Q}_D(X)^{-1} \qquad (X \ge X_0). \tag{2}$$

Here the finite region inequalities (17), (18), and (20) of [2] are used as external numerical inputs.

Proof. Fix ε first; in particular $\delta = \varepsilon^{1/3}$ is fixed. Whenever necessary, take ε smaller so that the deterministic sieve approximation errors fit strictly inside Li's positive margin. No accuracy parameter below changes δ, a moment order, or the number of Buchstab iterations.

Write $Q = \mathcal{Q}_D(X)$. We rerun the finite analytic estimates with an auxiliary accuracy Q^{-R}, where R is a sufficiently large fixed constant. This is not an application of Li's theorem with a moving B. The following checks show directly that its proof allows the new accuracy.

1. Pointwise input. Replace the low-frequency cutoff by $T_0 = Q^{R_0+2}$ with a sufficiently large fixed R_0. Prime factors satisfy every fixed power of Q^{-1} above that cutoff by Corollary 3.2 and Lemma 3.3. In particular, the saving Q^{-R_0} remains available after the fixed logarithmic losses. Below it, keep the main integral in Lemma 3.1. The contribution of its remainder is bounded by a fixed power of T_0 and $\log X$ times $e^{-a\mathcal{L}(X)}$, and is therefore smaller than any prescribed fixed power of Q^{-1}.

2. Mean and large values. The estimates in Li's Lemmas 3.1–3.6 use fixed moments and fixed interpolations of classical mean-value and large-value inequalities. In Lemma 3.1 the integer moment satisfies $k \ll 1/\delta$, and the logarithmic loss has exponent $O(\delta^{-2})$; both are fixed. Every resulting case has one of two forms:

$$(\log X)^d Q^{-\alpha R_0} \quad \text{or} \quad (\log X)^d X^{-\nu},$$

where d is fixed, $\alpha > 0$, and $\nu > 0$ depends on the fixed exponent margin. There are finitely many such cases. Thus the minimum relevant positive α can be chosen before R_0. By increasing R_0 and then X, every one is $O(Q^{-R})$ for any specified fixed R. The extra finite polynomial products in Lemmas 3.3 and 3.4 do not alter this conclusion. Lemmas 3.5 and 3.6 are regroupings and permutations of Lemma 3.2 and preserve the same accuracy accounting.

For completeness, one displayed interpolation in Lemma 3.2, case (1)(a), can be justified without its copied intermediate line. If the simultaneous large-value count F is bounded by

$$V^{-6}M^3, \quad W^{-10}H^5, \quad U^{-4}K^2, \qquad MHK = X,$$

interpolate with weights 3/10, 1/5, and 1/2. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2(V^{-6}M^3)^{3/10}(W^{-10}H^5)^{1/5}(U^{-4}K^2)^{1/2} \\ &= V^{1/5}M^{9/10}HK. \end{aligned}$$

If $V \le M^{1/2}Q^{-R_0}$, this is at most $XQ^{-R_0/5}$. The positive power of the pointwise saving is all that is needed. This is the interpolation already present in Jia's Lemma 5, case I(a), printed page 28.

3. The Watt-type estimates. Li's Section 4 uses a fixed exponent margin in its mean-value estimates. The resulting error is a fixed power saving in X, or one of the small-factor cases already covered in step 2. Both dominate every required fixed power of Q^{-1} by (1). The constants in these mean-value estimates depend on the fixed margins and coefficient bounds, not on a logarithmic accuracy exponent.

4. Perron and the exceptional measure. Here are the details behind the required error bookkeeping. For a relevant Dirichlet polynomial F, put

$$W_\eta(s) = \frac{(1+\eta)^s - 1}{s}.$$

On a dyadic range $|t| \asymp T$, $|W_\eta(b+it)| \ll \min(\eta, T^{-1})$, where $b = 1 + 1/\log X$. Opening the square, integrating $x^{s+\overline{s'}}$, and applying $2|ab| \le |a|^2 + |b|^2$, gives

$$\begin{aligned} &\int_X^{2X} \left| \int_{T_0 \le |t| \le X} F(b+it) W_\eta(b+it) x^{b+it}\, dt \right|^2 dx \\ &\qquad \ll X^3 (\log X)^3 \max_{T_0 \le T \le X} \min(\eta, T^{-1})^2 \int_{T \le |t| \le 2T} |F(b+it)|^2 dt. \end{aligned} \tag{3}$$

This is the estimate used in Jia's Lemmas 16–19 [5], whose Perron steps underlie Li's Section 5. If the right-hand maximum is $\ll \eta^2 Q^{-R}$, Chebyshev's inequality, with threshold $\eta X Q^{-r}$, yields exceptional measure

$$\ll X(\log X)^3 Q^{-R+2r}.$$

Choose $R > 2r + 10$ after all fixed decompositions. This leaves more than enough accuracy for (2). The low-frequency terms are evaluated as main terms, with errors $\ll \eta X Q^{-r}$ by step 1. Terms such as $\eta^2 X T_0^j$ and Perron truncation errors are harmless since $T_0^j = X^{o(1)}$ and ηX is a fixed positive power of X. Endpoint conventions contribute smaller errors as well. The same argument with the appropriate zeta factor handles the Type I terms.

5. Separation of the finite constraints. The number of prime variables and the number of defining inequalities are fixed. Standard dyadic partitions introduce only a fixed power of $\log X$. Alternatively, a relative mesh Q^{-r_1} separates the variables into at most $Q^{O_\varepsilon(r_1)}$ boxes. Boundary strips in fixed monomial and ordering constraints have total tuple weight $\ll XQ^{-r_1}(\log X)^{O_\varepsilon(1)}$; fixing all but one variable proves this bound by the elementary divisor estimate. Fubini and Markov transfer it to the short sums. Choose r_1 for the needed boundary accuracy first, and then choose R large enough to cover the number of boxes. This avoids a circular choice of accuracy powers. For the lower bound one may also shrink the admissible polyhedral regions by a sufficiently small fixed amount: their boundary loss is absorbed in the fixed positive numerical margin.

6. The sieve and its positive margin. In Li's Lemma 5.6 the sieve level $X^{10^{-1000}}$ and the sifting threshold X^{δ} are fixed powers of X. The bounded sieve coefficients and the grouping of finitely many variables preserve fixed divisor bounds. The deterministic errors of size $O(\varepsilon)$ in the normalized main term need not become Q^{-1}: choose ε small once. Only the fluctuating error and its exceptional set require the stronger accuracy.

In applying the small-factor estimates, a coefficient supported on rough integers is not automatically declared to have cancellation. The relevant groups contain prescribed prime factors, to which Lemma 3.3 applies after the remaining variables are fixed. If the group denoted N in Li's Lemma 4.4 consists only of the tiny sieve variable, use the Type I estimate in Lemma 5.1 instead: the relevant range has $QR < X^{45/86}$ and $N < X^{41/344}$. Thus the proof does not need a pointwise estimate for arbitrary sieve weights.

Li's recorded numerical region estimates give the positive margin

$$1 - 0.03 - 0.992 + 0.027 = 0.005.$$

Choose the fixed deterministic errors and region-boundary losses less than half this margin. Steps 1–5 make all remaining errors smaller than a further fixed portion outside measure XQ^{-1}, for sufficiently large X. The same fixed sieve decomposition therefore gives (2), with a positive constant κ. The numerical inequalities themselves are precisely the external component specified in the proposition. □

Remark 4.2 (what is and is not imported). The standard statement $O_B(X/\log^B X)$ is not the imported quantitative conclusion. We use the finite algebraic estimates and the fixed positive sieve margin, and verify their accuracy dependence. The general free-parameter variance estimate in Matomäki–Radziwiłł–Shao–Tao–Teräväinen [6, Lemma 3.3] provides a separate modern formulation of the Perron/variance step. Their other results are not used to claim a new short-interval exponent.

5 Transfer from short intervals to bad integer nodes

Proposition 5.1 (the published-source alternative). *For sufficiently small fixed $\varepsilon > 0$, the counterpart of Proposition 4.1 holds with $\eta = \frac{1}{2}X^{-19/20+\varepsilon}$, using Jia's published proof in place of Li's refined decomposition.*

Proof. Jia fixes ε, $\varepsilon_1 = \varepsilon^2$ and $\delta = \varepsilon^{1/3}$ before the accuracy parameter. His mean-value arguments in Sections 2–5 have the same structure as steps 1–3 in Proposition 4.1: fixed moments, fixed powers of logarithms, and either a positive power of the pointwise saving or a fixed power saving in X. His Lemmas 16–19 are the Perron calculations quantified directly in (3). The subsequent fixed Buchstab regions, linear sieve levels and deterministic $O(\varepsilon)$ or $O(\delta)$ errors do not depend on the requested logarithmic accuracy. Thus the identical order of parameter choices gives an exceptional measure at most $X\mathcal{Q}_D(X)^{-1}$ for every fixed D. Jia's published positive numerical sieve margin supplies the constant $\kappa > 0$. We use that published margin as an external input just as in a normal application of his theorem. □

A separate averaging argument avoids any need to force the sieve interval length to equal the length of $J_c(m)$.

Lemma 5.2 (longer interval averaging). *Suppose, for $u \in [X, 2X]$ outside a set E_X,*

$$\#(\mathcal{P} \cap (u, (1+\eta)u]) \geq \kappa \frac{\eta u}{\log X}.$$

Let $I = [a, b) \subset [X, 2X]$, $L = b - a$, and $L/(\eta X) \to \infty$. If $\#(\mathcal{P} \cap I) < \kappa L/(16 \log X)$, then

$$\operatorname{meas}(E_X \cap I) \geq L/4$$

for sufficiently large X.

Proof. Let $B_I = [a, b/(1+\eta))$. For large X, $\operatorname{meas} B_I \geq L/2$ and $(u, (1+\eta)u] \subset I$ for $u \in B_I$. Every prime of I is counted for at most $2\eta X$ units of u-measure. Consequently

$$\int_{B_I} \#(\mathcal{P} \cap (u, (1+\eta)u])\, du \leq 2\eta X \#(\mathcal{P} \cap I) < \frac{\kappa \eta X L}{8 \log X}.$$

On $B_I \setminus E_X$ the integrand is at least $\kappa\eta X/\log X$, so $\operatorname{meas}(B_I \setminus E_X) \leq L/8$. Thus $\operatorname{meas}(E_X \cap I) \geq 3L/8 \geq L/4$. □

Proposition 5.3 (bad-node estimate)**.** *Fix $43/41 < c_- \leq c_+ < \infty$. There is $\delta_0 > 0$ such that, for every $D > 0$, for sufficiently large Y, uniformly for $c \in [c_-, c_+]$,*

$$\#\left\{ m \in [Y, 2Y) : \#(\mathcal{P} \cap J_c(m)) < \delta_0 \frac{|J_c(m)|}{\log((m+1)^c)} \right\} \leq Y \exp(-D(\log\log Y)^2).$$

Proof. Choose a fixed sufficiently small $\varepsilon > 0$ with

$$\frac{2}{43} + \varepsilon < 1 - \frac{1}{c_-}.$$

At height $X \asymp Y^c$, the intervals $J_c(m)$ have length

$$|J_c(m)| \asymp Y^{c-1} = X^{1-1/c}.$$

This is uniformly larger by a fixed power of X than the intervals of length $\eta X \asymp X^{2/43+\varepsilon}$ in Proposition 4.1. The $J_c(m)$ are disjoint. Cover their heights with $O_{c_+}(1)$ dyadic ranges. At most $O_{c_+}(1)$ of the intervals cross a boundary; absorb these in the final estimate.

Choose δ_0 sufficiently small in terms of κ. By Lemma 5.2, each bad m accounts for $\gg Y^{c-1}$ measure of the exceptional set in its dyadic range. Apply Proposition 4.1 with a larger fixed accuracy constant, say $2D+1$, and divide the exceptional measure by Y^{c-1}. Since $\log\log(Y^c) = \log\log Y + \log c$, fixed factors and the dyadic cover are absorbed for large Y. This proves the estimate. □

6 The harmonic construction

We include the mass argument to specify the required quantifiers.

Lemma 6.1 (mass survival and binary subdivision)**.** *Let V_1 be finite, and let disjoint finite child sets $\mathrm{Ch}_n(m)$ define the successive levels $V_{n+1} = \bigcup_{m \in V_n \cap G_n} \mathrm{Ch}_n(m)$. Write $w(S) = \sum_{m \in S} 1/m$. Suppose*

$$w(\mathrm{Ch}_n(m)) \geq \lambda_n/m \quad (m \in V_n \cap G_n), \qquad w(V_n \setminus G_n) \leq \beta_n.$$

Put $s_n = \frac{1}{3} w(V_1) \prod_{j<n} (\lambda_j/6)$. If $\beta_n \leq s_n/2$ and $\max_{m \in V_n} 1/m \leq s_n$ for all n, there are uncountably many infinite branches.

Proof. A finite collection of total weight at least $3s_n$, with every atom of weight at most s_n, splits into two disjoint parts of weight at least s_n each, by a greedy cut. After selecting either part and discarding its bad nodes, its remaining weight is at least $s_n/2$. Its children have weight at least $\lambda_n s_n/2 = 3s_{n+1}$, so the operation repeats. For every infinite binary choice the selected forest has nodes at every depth. Local finiteness and the finite root set give an infinite path by König's lemma. Different binary choices have disjoint selected node sets at the first differing level, hence give different paths. □

Proof of Theorem 1.1, constant exponent. Fix $c > 43/41$ and a large P. Start from $V_1 = \mathcal{P} \cap [P, 2P]$ and put $\mathrm{Ch}_n(m) = \mathcal{P} \cap J_c(m)$. Take G_n to be the good integers in Proposition 5.3. Inductively,

$$V_n \subset [P^{c^{n-1}}, (3P)^{c^{n-1}}]$$

for large P. The upper bound follows by iterating $M_{n+1} = (M_n+1)^c$ and summing the increments $1/M_n$ after dividing logarithms by c^{n-1}.

The prime number theorem gives $w(V_1) \geq 1/(4\log P)$. Since $|J_c(m)| \sim cm^{c-1}$, goodness gives

$$w(\mathrm{Ch}_n(m)) \geq \frac{a}{m\, c^{n-1}\log(4P)}$$

with fixed $a > 0$. Set $\lambda_n = a/(c^{n-1}\log(4P))$. A dyadic cover of the displayed level range has $O(c^{n-1})$ pieces, so Proposition 5.3 yields

$$\beta_n \ll c^{n-1}\exp\left(-D\big(\log\log(P^{c^{n-1}})\big)^2\right).$$

Choose $D > 1/(2\log c)$. Write $m = n-1$, $d = \log c$, $\ell = \log P$, and $L = \log \ell$. For s_n in Lemma 6.1,

$$-\log s_n = \frac{d}{2}m^2 + mL + O(m+L+1),$$

whereas

$$\log \beta_n \leq -D(L+md)^2 + md + O(1).$$

Their difference is at most

$$-\left(Dd^2 - \frac{d}{2}\right)m^2 - (2Dd-1)mL - DL^2 + O(m+L+1).$$

Both bracketed coefficients are positive. A single sufficiently large P therefore ensures $\beta_n \leq s_n/2$ for every n. Also

$$\max_{p\in V_n}\frac{1}{p} \leq e^{-c^m\ell} \leq s_n$$

for all $m \geq 0$ once P is sufficiently large. Thus Lemma 6.1 supplies uncountably many infinite prime chains.

For such a chain, set

$$a_n = p_n^{1/c^n}, \qquad b_n = (p_n+1)^{1/c^n}.$$

The inequalities defining $J_c(p_n)$ imply $a_{n+1} \geq a_n$ and $b_{n+1} < b_n$. Hence $A = \lim a_n$ exists and $a_n \leq A \leq b_{n+1} < b_n$. Thus $\lfloor A^{c^n}\rfloor = p_n$ for every n. Different chains yield different A. Since P can be arbitrarily large, $\mathcal{W}(c)$ is unbounded.

Finally, fix $1 < a < b$. The set satisfying just the nth prime condition in $[a,b]$ is a finite union of intervals of total length at most

$$\frac{b}{c^n}\sum_{a^{c^n}-1\leq p\leq b^{c^n}}\frac{1}{p} = O_{a,b}(c^{-n}),$$

by Mertens' theorem. Its closure adds only finitely many endpoints. The closure of $\mathcal{W}(c)\cap[a,b]$ is contained in every such closed union, hence is null and has empty interior. This proves nullity and nowhere density. □

Bounded variable exponents. Let $D_n = c_2\cdots c_n$, $D_1 = 1$. Replace c^{n-1} above by D_n. The uniform estimate in Proposition 5.3 applies because $c_n \in [c_-, c_+]$. Use $\lambda_n = a/(D_n\log(4P))$ and choose

$$D > \frac{\log c_+}{2(\log c_-)^2}.$$

The same comparison, using $c_-^{n-1} \leq D_n \leq c_+^{n-1}$, proves the conditions of Lemma 6.1 uniformly at every level. The nested intervals now have exponents $1/C_n$. Their limits, distinctness, unboundedness, and the nullity argument are unchanged. □

Proof of Theorem 1.2. Use Proposition 5.1 in the sliding-window averaging argument. The only change in the bad-node and harmonic-tree proofs is the choice $1/20 + \varepsilon < 1 - 1/c_-$, which is possible exactly when $c_- > 20/19$. All other estimates are the same. □

7 What would lower the exponent further?

The extension above uses the first Li paper because its finite mean-value calculation and the Perron arguments behind it can be tracked with an explicit accuracy parameter. Li II states a lower-bound theorem at interval exponent $1/22$. Its additional ingredients include mean-value estimates quoted from Harman's monograph; the author-site manuscript Li III gives the further proposed exponent $1/24$. Both state fixed-B exceptional estimates.

The exact next task is to prove, for their additional sieve regions, the counterpart of Proposition 4.1:

$$\forall D > 0, \qquad \operatorname{meas}\{x : \text{the required lower bound fails}\} \ll X \exp(-D(\log\log X)^2).$$

If established at every exponent $\theta > 1/22$, it gives $c > 22/21$; at every $\theta > 1/24$, it gives $c > 24/23$. These are implications, not conclusions of the present proof.

The following distinctions should be preserved in any subsequent revision:

- A fixed finite prime chain is not evidence of an infinite chain.
- A new exceptional-set estimate must be proved with its parameter dependence; choosing B as a function of X in a fixed-B theorem is insufficient.
- The representing constant obtained by compactness need not be the limit of a specified greedy prime search.
- The endpoint $c = 43/41$ and the whole range $1 < c \leq 43/41$ remain outside the proved conclusion of this note.
- No historical-priority or publication claim is made.

Dependency record

1. The supplied harmonic branching manuscript [1] provides the original global-mass construction. Its core survival calculation is reproduced here.
2. Li I [2] supplies the finite admissible sieve regions, the large-value case structure, and the numerical integral bounds giving the positive margin 0.005.
3. Jia [5] supplies the underlying Perron and variance computations; their accuracy dependence is made explicit in (3).
4. Vinogradov–Korobov supplies the stronger pointwise error used in Lemma 3.1. All other losses in the audited finite argument are fixed powers or fixed positive powers of the new accuracy target.
5. The more recent manuscript thresholds are documented as research targets, not silently imported as already quantified theorems.

References

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